ContextBasedRAG

Simple guide: yeh software kya karta hai aur kaun sa page kab use karna hai.
Ek Mahindra plant ke example ke saath, step by step.

> **Dhyan do:** Mahindra sirf samjhane ke liye liya gaya example hai. Is guide ke log, machine aur values kalpanik (invented) hain. Software ka Mahindra company se koi sambandh nahi hai.

1. Ek line mein: yeh software kya hai

Yeh factory ke maintenance department ka **sawaal-jawab wala assistant** hai. Company apni machines ke manual, SOP aur purane breakdown ka record isme daalti hai. Phir koi bhi worker apni bhasha mein sawaal poochta hai, aur software **sirf company ke apne documents se** jawab deta hai, saath mein batata hai ki jawab kis file ke kis page se aaya.

2. Factory mein problem kya hai (software ke bina)

- Machine pe alarm aata hai. Operator ko uska matlab nahi pata. Manual 300 page ka hai aur office ki almari mein hai.
- Senior technician chhutti pe hai. Sirf usko yaad tha ki pichhli baar yahi alarm kaise theek hua tha.
- Manual ek value bolta hai, company ki SOP doosri. Worker ko nahi pata kaun si sahi hai.
- Repair ka kaam bina kisi ke check kiye shuru ho jaata hai. Baad mein koi record nahi ki kisne kya kiya.

Nateeja: machine zyada der band rehti hai (downtime), aur galat value se kaam karne ka khatra rehta hai.

3. Software in problems ko kaise hal karta hai

Problem	Software ka jawab	Kaun sa page
Manual dhoondhna mushkil	Sawaal poocho, jawab source ke saath milta hai	Ask
Purana fix kisi ko yaad nahi	Har breakdown ka record rehta hai aur jawab mein judta hai	Breakdowns, Ask
Do documents alag value bolte hain	Dono value dikhata hai, khud koi ek nahi chunta; ek zimmedar banda faisla likhta hai	Conflicts
Kaam bina check ke shuru	Work order pehle approve hota hai, tabhi kaam shuru hota hai	Work orders
Kisne kya kiya, record nahi	Har kaam apne aap likha jaata hai	Audit history

4. Example: Mahindra ka ek plastic moulding shop

Maan lo Mahindra ke ek plant mein gaadi ke plastic parts (bumper, dashboard ke hisse) bante hain. Wahan ek injection moulding machine hai jiska code **IM-150** hai. Is example mein chaar log hain:

Naam	Kaam	Software mein role	Kya kar sakta hai
Rajesh	Plant head	Administrator	Sab kuch. Company ka account wahi banata hai, users jodta hai, sabko access deta hai.
Sunita	Maintenance manager	Maintenance manager	Machines aur documents sambhalti hai, work order approve karti hai, conflict ka faisla karti hai.
Amit	Technician	Technician	Breakdown likhta hai, work order banata hai, approve hone ke baad kaam karta hai.
Ramesh	Machine operator	Viewer	Sirf dekh sakta hai aur sawaal pooch sakta hai.

5. Menu ka naksha: kaun sa page kis kaam ka

Page	Ek line mein kaam	Zyada kaun use karta hai
Dashboard	Poori factory ka haal ek nazar mein: kitni machines, kitne breakdown khule, kitne work order approval ke intezaar mein	Rajesh, Sunita
Ask	Sawaal poocho, documents se jawab lo	Sab log
Assets	Machines ki list. Har machine ka apna page	Sunita
Documents	Manual aur SOP upload karne ki jagah	Sunita
Breakdowns	Machine kab kharab hui, kyun, kaise theek hui: iska register	Amit
Work orders	Repair ka likhit order: banao, approve karao, kaam karo, band karo	Amit, Sunita
Conflicts	Jahan do documents alag value bolte hain, uski list aur faisla	Sunita
Audit history	Kisne kab kya kiya, uska poora record	Rajesh, Sunita
Users	Log jodna aur har ek ka access tay karna	Sirf Rajesh

6. Pehli baar setup (ek hi baar karna hai)

Yeh kaam shuru mein Rajesh aur Sunita karte hain. Iske baad roz ka kaam section 7 jaisa chalta hai.

> **Step 1. Company ka account banao.** Rajesh
> Website kholo, "Register a factory" chuno, company ka naam, apna naam, email aur password bharo. Jo account banata hai wahi Administrator ban jaata hai.

> **Step 2. Apni team jodo.** Rajesh - page: **Users**
> "Add user" dabao. Sunita ko role Maintenance manager, Amit ko Technician, Ramesh ko Viewer do. Har ek ko uska email aur shuruaati password khud batao.

> **Step 3. Machines darj karo.** Sunita - page: **Assets**
> "Add asset" dabao. Code: IM-150, naam: Injection moulding machine, location: Moulding shop, criticality: high. Plant ki har machine ke liye yahi karo.

Step 4. Documents upload karo. Sunita - page: **Documents**
"Upload document" dabao, IM-150 ka manual (PDF) chuno, Type mein "manual" aur "Applies to assets" mein IM-150 chuno. Company ki SOP bhi isi tarah upload karo, Type "sop". Status "Ready" hone ka matlab software ne file padh li.

Step 5. Purane breakdown daalo (agar Excel mein hain). Sunita - page: **Breakdowns**
"Import CSV / XLSX" se purana register ek saath aa jaata hai. Isse software ko machine ka itihaas mil jaata hai.

Step 6. Zaroorat ho toh kisi ka access badlo. Rajesh - page: **Users**
Kisi user ke saamne "Access" dabao aur tick lagao ki usko kaun sa section dikhe aur woh kya kar sake. Example: Amit senior technician hai, toh usko "Approve work orders" ka tick de sakte ho.

7. Roz ka kaam: ek breakdown ki poori kahani

Subah 10 baje IM-150 pe **alarm A-207** aata hai aur machine ruk jaati hai. Ab dekho kaun kya karta hai.

Step 1. Operator sawaal poochta hai. Ramesh - page: **Ask**
Ramesh likhta hai: "IM-150 pe alarm A-207 ka matlab kya hai aur kya check karna hai?"
Software jawab deta hai: barrel temperature low hai, heater band HB-3 aur thermocouple TC-J2 check karo. Neeche source likha hota hai: manual, page 4. Ramesh file kholke khud dekh sakta hai.

Step 2. Technician breakdown darj karta hai. Amit - page: **Breakdowns**
"Report breakdown" dabata hai. Asset: IM-150, error code: A-207, symptoms: "Barrel temperature low, machine ruk gayi". Ab yeh record hamesha ke liye machine ke itihaas mein hai.

Step 3. Technician work order banata hai. Amit - page: **Assets** (IM-150) ya **Work orders**
"Draft work order" dabata hai aur problem likhta hai. "Generate from records" chunne par software manual, SOP aur purane breakdown se kaam ke steps ki list banata hai, har step ke saath source. Amit steps padhta hai, zaroorat ho toh badalta hai, phir "Submit for approval" dabata hai. Submit ke baad order badla nahi ja sakta.

Step 4. Manager check karke approve karti hai. Sunita - page: **Work orders**
"Approval queue" tab mein order dikhta hai. Sunita steps aur unke source dekhti hai. Sahi hai toh "Approve", galat hai toh "Reject" (wajah likhna zaroori hai). Dashboard pe bhi "awaiting approval" ka number dikhta hai.

Step 5. Kaam hota hai. Amit - page: **Work orders**
Approve hone ke baad Amit "Start work" dabata hai, heater band HB-3 badalta hai, phir "Mark completed" dabata hai.

Step 6. Breakdown band karo. Amit - page: **Breakdowns**
Us record pe "Update" dabata hai: root cause "heater band HB-3 jal gaya", corrective action "heater band badla", parts used "HB-3", downtime 90 minutes, status "resolved".

Step 7. Agli baar fayda. Koi bhi - page: **Ask**
Teen mahine baad wahi alarm phir aata hai. Naya operator poochta hai: "IM-150 pe A-207 pehle aaya tha kya?" Software manual ke saath pichhla breakdown record bhi dikhata hai: pichhli baar HB-3 badalne se theek hua tha. Yahi is software ka asli fayda hai: anubhav kisi ek aadmi ke dimaag mein nahi, system mein rehta hai.

8. Har page detail mein

Dashboard

Kya dikhta hai: machines ki ginti, kitne documents ready hain, kitne fail hue, kitne breakdown abhi khule hain, kitne work order approval ke intezaar mein hain, aur kis machine ka downtime sabse zyada hai (chart). Manager ko yeh bhi dikhta hai ki kaun se sawaalon ka jawab records mein nahi mila; isse pata chalta hai kaun sa manual upload karna baaki hai.

Example: Rajesh subah Dashboard kholta hai aur dekhta hai ki IM-150 ka downtime is mahine sabse zyada hai.

Ask

Kya karta hai: sawaal ka jawab sirf upload kiye gaye documents aur breakdown records se deta hai. Hindi, English ya Hinglish, kisi mein bhi pooch sakte ho.

- Jawab ke neeche hamesha source hota hai: file ka naam aur page. Us pe click karke asli file khulti hai.
- Sawaal mein machine ka code likho (jaise IM-150) toh software sirf us machine ke documents dekhta hai.
- Agar jawab documents mein nahi hai toh software saaf bolta hai "No supporting evidence found". Woh andaaza nahi lagata. Yeh galti nahi, suraksha hai.

Assets

Kya hai: factory ki machines ka register. Kisi machine pe click karo toh uska apna page khulta hai jahan ek hi jagah uski details, uske documents, breakdown ka saar (kaun sa alarm kitni baar aaya, kul downtime) aur uske work orders dikhte hain.

Kyun zaroori: asset hi woh cheez hai jisse documents, breakdowns aur work orders jude hote hain. Bina asset ke breakdown ya work order nahi ban sakta.

Documents

Kya hai: manual, SOP aur spare parts list upload karne ki jagah (PDF, Word, PowerPoint, text).

- **Manual** machine banane wali company ka hota hai. **SOP** factory ka apna likha tarika hota hai.
- Upload ke waqt batao yeh kis machine ka hai. Kuch na chuno toh document "General" rehta hai, yaani sab machines ke liye.
- Status: Queued / Processing (padha ja raha hai), Ready (sawaal pooch sakte ho), Failed (dobara "Reprocess" dabao).

Breakdowns

Kya hai: machine kharab hone ka register: kab, kaun si machine, kya lakshan, error code, kitni der band rahi, asli wajah, kya theek kiya, kaun sa part laga.

Kyun zaroori: yahi record baad mein Ask ke jawab aur work order ke steps mein kaam aata hai. Jitna achha record, utna achha jawab.

Work orders

Kya hai: repair ke kaam ka likhit order. Iska safar hamesha is kram mein chalta hai:

Draft (banaya) > **Pending approval** (submit kiya) > **Approved** (manzoor) > **In progress** (kaam shuru) > **Completed** (kaam khatam)

- Software ya AI khud kuch approve nahi karta. Approve hamesha insaan karta hai.
- Jis step ka koi source nahi hai ya jo safety se juda hai (bijli, hydraulic pressure, lockout), us pe nishaan lagta hai taaki approve karne wala dhyan se dekhe.

- Apna banaya order khud approve nahi kar sakte; doosra banda chahiye. Administrator is niyam se bahar hai, aur woh chahe toh kisi aur ko bhi yeh chhoot de sakta hai.

Conflicts

Kya hai: jab do documents ek hi cheez ke liye alag value bolte hain, uski list.

> **Example:** IM-150 ka manual kehta hai hydraulic oil ka temperature 40 se 55 degrees C hona chahiye. Company ki SOP kehti hai 40 se 65. Ramesh poochta hai "oil temperature kitna hona chahiye?"
> Software 55 ya 65 mein se koi ek nahi chunta. Woh dono value, dono file ke naam ke saath dikhata hai aur chetavni deta hai. Yeh entry Conflicts page pe aa jaati hai. Sunita "Record a decision" dabake likhti hai ki manual sahi hai aur SOP sudhaarni hai. Uske baad sabko yahi faisla dikhta hai.

Kyun zaroori: galat temperature se machine ya aadmi ko nuksaan ho sakta hai. Aam chatbot chupchaap koi ek value bol deta; yeh software aisa nahi karta.

Audit history

Kya hai: har kaam ka apne aap bana record: kisne asset banaya, kisne document upload ya delete kiya, kisne work order approve kiya, kiski koshish mana hui. Software mein ise badalne ya mitane ka koi button nahi hai.

Example: audit ke waqt Rajesh dikha sakta hai ki IM-150 ka repair kisne approve kiya aur kab.

Users

Kya hai: sirf Administrator ke liye. Yahan log jodte hain, role badalte hain, kisi ko band (Deactivate) karte hain, aur "Access" se har ek ke liye tay karte hain ki kaun sa section dikhega aur woh kya kar sakta hai.

9. Kaun kya kar sakta hai (role ka standard access)

Kaam	Viewer	Technician	Manager	Admin
Dekhna aur sawaal poochna	Haan	Haan	Haan	Haan
Breakdown likhna / update karna	-	Haan	Haan	Haan
Work order banana, kaam shuru / khatam karna	-	Haan	Haan	Haan
Machines aur documents sambhalna	-	-	Haan	Haan
Work order approve / reject karna	-	-	Haan	Haan
Conflict ka faisla likhna	-	-	Haan	Haan
Audit history dekhna	-	-	Haan	Haan
Apna banaya work order khud approve karna	-	-	-	Haan
Users jodna aur access dena	-	-	-	Haan

Yeh standard hai. Administrator "Users > Access" se kisi ek user ke liye isme se kuch bhi jod ya hata sakta hai (sirf "Users jodna" kisi aur ko nahi diya ja sakta).

10. Software ke chaar pakke niyam

1. **Andaaza nahi.** Jawab documents mein nahi hai toh software bolta hai ki nahi mila.
2. **Har jawab ka source.** File ka naam aur page hamesha saath mein hota hai.
3. **Conflict mein khud faisla nahi.** Do value hain toh dono dikhata hai; faisla insaan karta hai.
4. **AI sirf draft likhta hai.** Approve aur kaam hamesha insaan karta hai, aur sab kuch audit mein darj hota hai.

Ek aur baat: har company ka data alag rehta hai. Ek factory ke log doosri factory ka kuch nahi dekh sakte.

11. Project dikhane ke liye 5 minute ka demo

1. **Documents:** dikhao ki IM-150 ka manual aur SOP upload hain aur Ready hain.
2. **Ask:** poocho "IM-150 pe alarm A-207 ka matlab kya hai?" Jawab aur uska source (page) dikhao.
3. **Ask:** poocho "IM-150 ka mould temperature kitna rakhna hai?" Software bolega records mein nahi hai. Batao ki yeh andaaza nahi lagata.
4. **Ask:** poocho "IM-150 ka hydraulic oil temperature kitna hona chahiye?" Conflict ki chetavni dikhao (55 aur 65).
5. **Conflicts:** "Record a decision" se faisla likho.
6. **Assets > IM-150:** "Draft work order" se AI wala draft banao, steps aur source dikhao.
7. **Work orders:** Submit, Approve, Start work, Mark completed.
8. **Users:** "Access" kholke dikhao ki har user ka access alag tay ho sakta hai.
9. **Audit history:** dikhao ki upar ka har kaam darj hua hai.

12. Aam dikkatein aur unka matlab

Kya dikha	Matlab	Kya karna hai
"No supporting evidence found"	Jawab upload kiye documents mein nahi hai	Sahi manual ya SOP upload karo, ya sawaal badlo
Document ka status "Failed"	File padhi nahi ja saki ya AI key mein dikkat	Error padho, phir "Reprocess" dabao
Approve button nahi dikh raha	Order tumne khud banaya hai, ya tumhe approve ka access nahi hai	Administrator se approve karao ya access maango
Koi section menu mein nahi hai	Administrator ne woh section tumhare liye band rakha hai	Administrator se access maango
"Request quota is used up"	Free AI key ka roz ka limit (lagbhag 20 AI jawab) khatam	Agle din try karo. Conflict wale jawab is limit mein nahi ginte